Global eligible pairs behind a six-point head

Human Compression project: L044 continuation

Retain the full standing assumptions and smaller-degree alternating induction of the accepted critical-head and paired-head theorems. In particular $G \le A_n$ is transitive with a connected string C-marking, $n \ge 12$, and every maximal ambient omission is intransitive. The first three generators form the actual supported six-point head of source Table 2(2) or (3), in an extendable orientation. The complete exceptional-head interface is required: on $D = \{1, \dots, 6\}$ the head is transitive; $\tau = \rho_3 = (6\ b)\delta$, $\sigma = \rho_4$ fixes D, $H = G_{\ge 5}$ fixes $D \cup \{b\}$, and $B = \langle \sigma, H \rangle$ is transitive on $\Omega \setminus D$. Under rank failure the accepted reduction supplies $n = 2r \ge 18$ and the actual natural alternating tail used in the pure-head proof. Write $K = G_{>5}$. That proof gives

$$x = b^\sigma \ne b, \qquad x^\tau = x, \qquad K \text{ fixes } x. \tag{1}$$

It uses full ambient IP, not a quotient IP.

Theorem 1. *If every colour $j \ge 5$ has at least two ambient-eligible edges anywhere in Ω, then $r \le \lfloor (n-1)/2 \rfloor$. Consequently a counterexample in this exact head configuration has a further colour $j \ge 5$ with exactly one globally ambient-eligible edge.*

Proof. Suppose the bound fails. Let f be the number of H-fixed points and c its number of nontrivial orbits. The full ambient-interval bound gives $f + c \le 10$, while $f \ge 7$ and $c \ge 1$. If $c = 1$, every eligible pair is already in that single nontrivial orbit, so the complete paired-head theorem applies.

Suppose $c = 2$. Then $f = 7$ or 8, and τ fixes all additional H-fixed points. Its action outside the seven-point head is odd. If it preserves both nontrivial orbits, its nontrivial actions there are free, and it is free on at least one orbit. Any eligible edge on a free orbit has a distinct eligible translate by τ: an invariant unordered edge would be a τ-pair and hence ineligible since $\tau \in G_j$. Thus every colour with an edge on a free orbit has a pair there; a colour with no such edge has its two edges in the single remaining fixed orbit. The paired-head theorem again applies.

If τ exchanges the two orbits, they have the same odd size. Their union has even size, so $f = 8$. By (1), x cannot lie in either exchanged orbit, nor in $D \cup \{b\}$. It is therefore the sole additional H-fixed point. Both H and σ preserve $\{b, x\}$, contradicting transitivity of B. Here x is first known to be K-fixed; its being H-fixed follows only from this exact orbit count and $\tau x = x$.

It remains that $c = 3, f = 7$. The nontrivial support has odd size $n - 7$. If τ preserves all three orbits and is free on at least two, then every colour has a pair in one orbit: any edge on a free orbit supplies that pair, and at most one fixed orbit remains. It cannot be free on all three or act trivially on all three. Thus the only preserved case not already covered has an odd free action on one orbit O of size $2q$, with q odd, and fixes the other two orbits pointwise. Call their union V and put $v = |V|$; then v is odd and

$$r - 5 = q + (v - 3)/2. \tag{2}$$

Partition all tail labels into L, those with any ambient-eligible edge in O, and F, those with none. These sets are disjoint and exhaust the tail. Choose one such edge for each label in L and project to the q pairs of τ in O. Each is a genuine edge between different pairs. The projected edges form a forest: a cycle would lift to an alternate path avoiding one label, using

the other selected edges and $\tau \in G_j$ inside pairs. This contradicts ambient eligibility of that label. Hence $|L| \leq q-1$. Every eligible edge of a colour in F lies in V, since the head is fixed. The selected-localization theorem, using the full tail interval and its two actual orbits in V, gives $2|F| \leq v-2$, and hence $|F| \leq (v-3)/2$. Together these bounds contradict (2), including $q=1$.

The final possibility is that τ exchanges two nontrivial orbits O, O' of odd size $m \geq 3$ and fixes the remaining odd orbit V of size $v \geq 3$. Then

$$r - 5 = m + (v-3)/2. \tag{3}$$

Let L be the labels with an eligible edge in O; equivariance makes these exactly the labels with an eligible edge in O'. Put $F = \{5, \ldots, r-1\} \setminus L$. One chosen edge per L label is a forest on O, so $|L| \leq m-1$. All eligible F-edges lie in V, and the selected-localization bound gives $|F| \leq (v-1)/2$. Equality is forced in both bounds by (3).

The $m-1$ distinguished L restrictions to O are independent: if one were generated by the others, its selected edge would lie within an ambient omission orbit. The retained Whiston equality and complete odd-degree maximal-string theorem identify their marked image with the ordinary transposition path of S_m. For $m=3$ this is simply the pair of distinct transpositions in S_3; the accepted theorem covers all odd $m \geq 5$, including its degree-five certificate. Connected noncommutations force L to be consecutive in the original string.

Deleting one path label leaves its two path-component orbits. Every F restriction preserves these two sets: otherwise adjoining it to the other path generators makes the row transitive, contrary to the chosen eligible edge. Preservation of all path cuts forces every F label to fix the row pointwise, and likewise the exchanged row O'. Hence G_F is actually supported on V. The complement of the consecutive L interval has no one-colour hole, because $|L| = m-1 \geq 2$. Apply the second selected-localization theorem to G_F and V:

$$v - 1 = 2|F| \leq v - o_{G_F}(V).$$

Thus G_F is transitive on V. This conclusion concerns the actual supported subgroup; no projected intersection property was used to obtain it.

If the first tail label 5 belongs to L, then $G_F \leq K$. It centralizes σ and has just one nontrivial orbit, namely V, so σ preserves V. This contradicts transitivity of B. Suppose instead $5 \in F$. By (1) and the exchanged rows, $x \in V$. The set F consists of an initial interval and possibly a terminal interval. Its terminal subgroup lies in K, fixes x, and commutes with the initial subgroup. Because their product is transitive on V, the terminal subgroup fixes every point of V. It is supported there, so is trivial; nonidentity of every actual marked generator forces that terminal interval to be empty.

Consequently L is terminal. Its last label λ is at distance at least two from the initial F interval, since $|L| \geq 2$. It centralizes G_F and fixes x by membership in K, so fixes V pointwise. But all of L belongs to K and is transitive on each row O, O', and λ acts nontrivially on each. These two rows are therefore actual K-orbits on which λ acts nontrivially, whereas it fixes every point outside them. The permutation σ centralizes K, so transports its orbits equivariantly and commutes with λ. It must preserve $O \cup O'$. The group H also preserves this union, contradicting transitivity of B for the final time.

These cases exhaust the odd action of τ outside its head. Every tail colour has at least one ambient-eligible edge by full transitivity and intransitivity of its omission. The contrapositive therefore gives the stated unique edge. □

Attribution and remaining boundary. This completes the global-pair version of the exceptional-head step supporting Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.31, physical pages 42–44. The source's orbit-deficit strategy and all accepted head, normal-form, pure-group and finite maximal-string inputs remain paid. The new selected-localization bounds and equality routing replace an

assumption that both edges were already in one orbit. The fixed-point distinction in the exchanged two-orbit case is explicit. The remaining unique-crossing route, stronger source equality diagrams, tables and whole L044 recovery are not claimed. No new census, formal declaration, original credit or question solution is asserted.